

NAME: _____

TEACHER: _____



The Scots College

Year 12 HSC Mathematics Extension 2

Assessment 3

June 2013

General Instructions

- Working time - 45 minutes
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- A table of standard integrals is provided at the end of the paper

TOTAL MARKS: 36

WEIGHTING: 20 %

Question	Topic	Max Marks	Marks Obtained
1	Integration	24	
2	Volumes	12	
Total		36	

Question 1 (24 Marks)

a) Find the following integrals

(i) $\int \frac{x-1}{\sqrt{1-x^2}} dx$ [2]

(ii) $\int \frac{e^{4x}-1}{e^x+1} dx$ [3]

(iii) $\int \frac{dx}{3 \sin x + 4 \cos x}$ [3]

(iv) $\int x \tan^{-1} x dx$ [3]

b) (i) Express $\frac{x^3}{(x-1)(x-2)}$ in the form $ax + b + \frac{c}{x-1} + \frac{d}{x-2}$ where a, b, c and d are real numbers.

(ii) Hence evaluate $\int_3^4 \frac{x^3}{(x-1)(x-2)} dx$ [4]

c) (i) Let $I_n = \int_0^1 x^n \sqrt{1-x} dx$ where n is a non-negative integer.

Show that $I_n = \frac{2n}{2n+3} I_{n-1}$, for $n \geq 1$.

(ii) Hence evaluate $\int_0^1 x^3 \sqrt{1-x} dx$. [5]

d) (i) Show that $\int_{-a}^a f(x) dx = \int_0^a [f(x) + f(-x)] dx$.

(ii) Hence evaluate $\int_{-\pi}^{\pi} \frac{\cos^2 x}{1+e^x} dx$ [4]

.....Question 2 on next page

Question 2 (12 Marks)

a) A solid sphere is formed by rotating the circle $x^2 + y^2 = 36$ about the y - axis. A cylindrical hole of diameter 6 cm is bored through the centre of the hole in the direction Oy .

(i) By considering a slice perpendicular to the y – axis, use the method of cylindrical shell to determine the volume of the solid remaining.

(ii) Also determine the volume of the section cut out from the sphere.

[6]

b) The region bound by the curve $y = x(4 - x)$ and the x – axis is rotated about the y – axis.

(i) By considering the solid formed by taking a slice perpendicular to the y – axis, show that the volume δV of this solid is given as $8\pi\sqrt{4 - y}$ cubic units.

(ii) Hence determine the volume of the solid of revolution.

[6]

... END OF EXAMINATION ...

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

Yr 12 HSC MATHEMATICS EXTENSION 2

2.07.8m

ASSESSMENT 3 - SOLUTIONS - JUNE 2013

$$1 \text{ (a)} \int \frac{x-1}{\sqrt{1-x^2}} dx$$

(i)

$$= \int \left(\frac{x}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x^2}} \right) dx$$

$$= -\frac{1}{2} \frac{(1-x^2)^{1/2}}{1/2} - \sin^{-1} x + C$$

$$= -\sqrt{1-x^2} - \sin^{-1} x + C$$

$$(iii) \int \frac{dx}{3\sin x + 4\cos x}, \quad \text{let } t = \tan \frac{x}{2}$$

$$dx = \frac{1}{2} \sec^2 \frac{x}{2} dx$$

$$= \int \frac{2dt/(1+t^2)}{3\left(\frac{2t}{1+t^2}\right) + \frac{4(1-t^2)}{1+t^2}}$$

$$dx = \frac{2dt}{1+t^2}$$

$$= \int \frac{2dt/(1+t^2)}{(6t + 4 - 4t^2)/(1+t^2)}$$

$$= \int \frac{dt}{2 + 3t - 2t^2}$$

$$= \int \frac{dt}{(2-t)(1+2t)}$$

$$\text{let } \frac{1}{(2-t)(1+2t)} = \frac{A}{2-t} + \frac{B}{1+2t}$$

$$1 = A(1+2t) + B(2-t)$$

$$= \int \left(\frac{1}{5(2-t)} + \frac{2}{5(1+2t)} \right) dt$$

$$t = 2 \quad 1 = 5A \quad A = \frac{1}{5}$$

$$t = -\frac{1}{2} \quad 1 = \frac{5B}{2} \quad B = \frac{2}{5}$$

$$= \frac{1}{5} \left[-\log(2-t) + \frac{2}{2} \log(1+2t) \right] + C$$

$$= \frac{1}{5} \log \frac{1+2t}{2-t} + C$$

$$= \frac{1}{5} \log \frac{1 + 2 \tan \frac{x}{2}}{2 - \tan \frac{x}{2}} + C$$

$$(iii) \int x \tan^{-1} x \, dx \quad u = \tan^{-1} x \quad v' = x$$

$$u' = \frac{1}{1+x^2} \quad v = \frac{x^2}{2}$$

$$= \frac{x^2}{2} \tan^{-1} x - \int \frac{x^2}{2(1+x^2)} \, dx$$

$$= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \left(\frac{x^2+1-1}{1+x^2} \right) \, dx$$

$$= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \left(1 - \frac{1}{1+x^2} \right) \, dx$$

$$= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} [x - \tan^{-1} x] + C$$

$$= \frac{x^2}{2} \tan^{-1} x + \frac{1}{2} \tan^{-1} x - \frac{1}{2} x + C$$

$$(b) (i) I_n = \int_0^1 x^n \sqrt{1-x} dx.$$

$$I_n = \left[\frac{2}{3} x^n (1-x)^{3/2} \right]_0^1 + \frac{2n}{3} \int_0^1 (1-x) x^{n-1} dx \quad \begin{matrix} u = x^n & v' = (1-x)^{1/2} \\ u' = nx^{n-1} & v = \frac{2}{3} (1-x)^{3/2} \end{matrix}$$

$$= [0-0] + \frac{2n}{3} \int_0^1 (1-x) \sqrt{1-x} x^{n-1} dx$$

$$= \frac{2n}{3} \int_0^1 [x^{n-1} \sqrt{1-x} - x^n \sqrt{1-x}] dx$$

$$= \frac{2n}{3} I_{n-1} - \frac{2n}{3} I_n$$

$$I_n \left(1 + \frac{2n}{3} \right) = \frac{2n}{3} I_{n-1}$$

$$I_n \left(\frac{3+2n}{3} \right) = \frac{2n}{3} I_{n-1}$$

$$\therefore I_n = \frac{2n}{2n+3} I_{n-1}$$

$$(ii) \int_0^1 x^3 \sqrt{1-x} dx = I_3 = \frac{6}{9} I_2$$

$$= \frac{6}{9} \left[\frac{4}{7} \left\{ \frac{2}{5} I_0 \right\} \right]$$

$$I_0 = \int_0^1 \sqrt{1-x} dx = \left[-\frac{2}{3} (1-x)^{3/2} \right]_0^1$$

$$= 0 - \left(-\frac{2}{3} \right) = \frac{2}{3}$$

$$\therefore I_3 = \frac{6}{9} \times \frac{4}{7} \times \frac{2}{5} \times \frac{2}{3} = \frac{96}{945}$$

$$(c) (i) \int_{-a}^a f(x) dx = \int_{-a}^0 f(x) dx + \int_0^a f(x) dx$$

$$\text{let } I_1 = \int_{-a}^0 f(x) dx \quad \text{let } y = -x \quad \frac{dy}{dx} = -1 \quad \text{or } -dy = dx$$

When $x = 0, y = 0$
 $x = -a, y = a$

$$I_1 = \int_a^0 f(-y) (-dy)$$

$$= \int_0^a f(-y) dy$$

$$= \int_0^a f(-x) dx$$

$$\therefore \int_{-a}^a f(x) dx = \int_0^a f(-x) dx + \int_0^a f(x) dx$$

$$= \int_0^a [f(x) + f(-x)] dx$$

$$(ii) \int_{-\pi}^{\pi} \frac{\cos^2 x}{1+e^x} dx = \int_0^{\pi} \left[\frac{\cos^2 x}{1+e^x} + \frac{[\cos(-x)]^2}{1+e^{-x}} \right] dx$$

$$= \int_0^{\pi} \left[\frac{\cos^2 x}{1+e^x} + \frac{\cos^2 x}{1+\frac{1}{e^x}} \right] dx$$

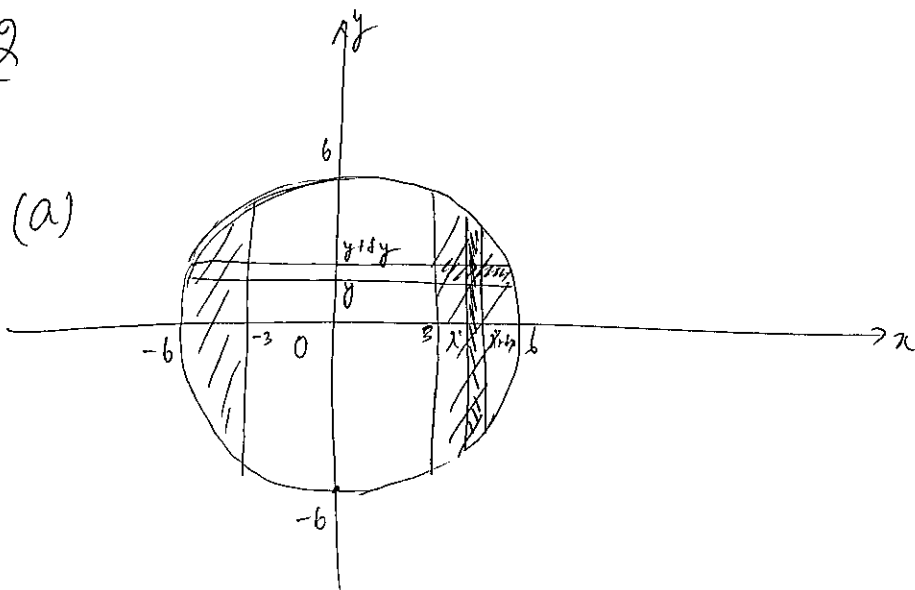
$$= \int_0^{\pi} \left(\frac{\cos^2 x}{1+e^x} + \frac{e^x \cos^2 x}{e^x + 1} \right) dx$$

$$= \int_0^{\pi} \frac{\cos^2 x (1+e^x)}{1+e^x} dx = \int_0^{\pi} \cos^2 x dx$$

$$= \frac{1}{2} \int_0^{\pi} (1 + \cos 2x) dx = \frac{1}{2} \left[x + \frac{1}{2} \sin 2x \right]_0^{\pi} = \frac{1}{2} [\pi + 0 - 0 - 0] = \frac{\pi}{2}$$

9.59.

Q2



$$dV = \pi [(x+dx)^2 - x^2] \cdot 2y$$

$$= \pi [x^2 + 2x dx + (dx)^2 - x^2] \cdot 2y$$

$$= \pi [2x dx] 2y$$

$$V = 4\pi \int_3^6 xy \, dy$$

$$= 4\pi \int_3^6 x \sqrt{36-x^2} \, dx$$

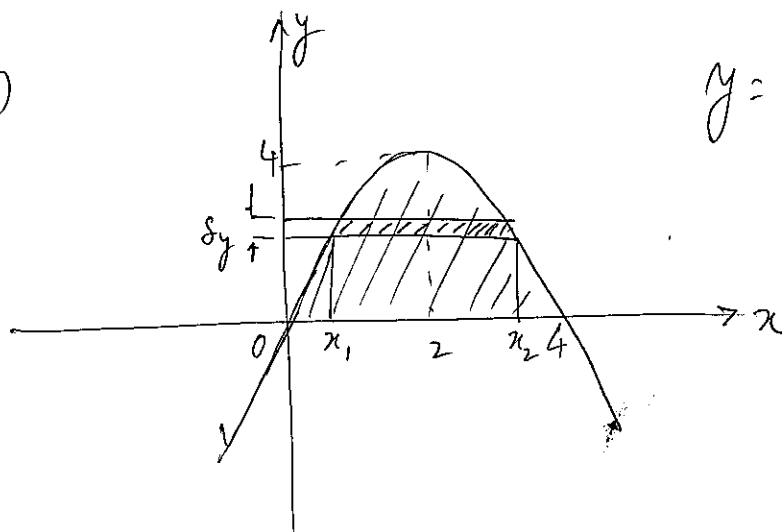
$$= 4\pi \left[-\frac{1}{x} \frac{(36-x^2)^{3/2}}{3/2} \right]_3^6$$

$$= 4\pi \left[0 + \frac{1}{3} (36-9)^{3/2} \right]$$

$$= 4\pi \left[0 + \frac{1}{3} 25^{3/2} \right]$$

$$= \frac{500\pi}{3} \text{ u}^3$$

(b)



$$y = x(4-x)$$

$$\delta V = \pi [x_2^2 - x_1^2] \delta y$$

$$= \pi [x_2 - x_1] [x_2 + x_1] \delta y$$

$$= \pi [4 (2\sqrt{4-y})] \delta y$$

$$= 8\pi \sqrt{4-y} \delta y$$

$$y = 4x - x^2$$

$$x^2 - 4x + y = 0$$

$$x = \frac{4 \pm \sqrt{16-4y}}{2}$$

$$= \frac{4 \pm 2\sqrt{4-y}}{2}$$

$$= 2 \pm \sqrt{4-y}$$

$$x_1 = 2 - \sqrt{4-y}$$

$$x_2 = 2 + \sqrt{4-y}$$

$$(ii) V = 8\pi \int_0^4 (4-y)^{1/2} dy$$

$$= 8\pi \left[\frac{-2(4-y)^{3/2}}{3} \right]_0^4$$

$$= 8\pi \left[-\frac{2}{3}(0) + \frac{2}{3} 4^{3/2} \right]$$

$$= \frac{8\pi}{3} \left(\frac{16}{3} \right) = \frac{128\pi}{9} u^3$$